

Codifying Chaos: A Rule-Based Approach to Procedural Generation of “Itano Circus” Animation

Anbang Wang*

Southwest Petroleum University
Chengdu, Sichuan, China
2768483241@qq.com

Tianyu Wang

Southwest Petroleum University
Chengdu, Sichuan, China
3517424084@qq.com

Abstract

Generating highly stylized animations without data-driven methods remains challenging. Current approaches like deep learning lack controllability, while physics-based simulations lack expressiveness, making styles like the “Itano Circus” difficult to reproduce. We propose a lightweight, rule-based method with a layered strategy: 1) Parametric Bézier curves ensure controllable macro-trajectories. 2) Envelope-modulated Simplex noise adds natural micro-wobble. 3) The Boids model enables emergent swarm behavior. Experiments validate the effectiveness and contributions of each component. Our code is available at <https://github.com/PapersAuthors/Codifying>.

CCS Concepts

• Computing methodologies → Motion processing.

Keywords

Layered Strategy, Simplex Noise, Boids Flocking Model, Itano Circus.

ACM Reference Format:

Anbang Wang and Tianyu Wang. 2026. Codifying Chaos: A Rule-Based Approach to Procedural Generation of “Itano Circus” Animation. In *The 2nd International Conference on Digital Society, Information Science and Risk Management (ICDIR 2026)*, January 23–25, 2026, Shenyang, China. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3804504.3804532>

1 Introduction

The procedural generation of highly stylized animation presents a fundamental challenge: translating artistic expression into algorithmic rules without sacrificing aesthetic authenticity. While deep learning approaches have advanced animation realism, they require substantial computational resources and lack the fine-grained control necessary for replicating deliberate artistic styles. This gap is acutely observed in the “Itano Circus” style, whose aesthetic presents a complex multi-agent motion problem, demanding the simultaneous synthesis of: (a) smooth, arcing macro-trajectories, (b) irregular, high-frequency micro-oscillations, and (c) dense, collision-free swarm coordination. [7].

Existing methods fall short. Physics simulations produce overly predictable motion. Standard particle systems lack Micro-behavior.

*Corresponding author



This work is licensed under a Creative Commons Attribution 4.0 International License. *ICDIR 2026, Shenyang, China*
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2288-2/26/01
<https://doi.org/10.1145/3804504.3804532>

Even sophisticated flocking algorithms fail to generate the characteristic acrobatic individual trajectories. Machine learning models, while powerful, struggle with the nuanced choreography and spatial reasoning inherent to this aesthetic, and their “black box” nature contradicts the goals of interpretability and artistic control.

As shown in Figure 1, To address these limitations, we formulate the problem through a hybrid agent-based model. Our solution decomposes the motion synthesis task into three decoupled algorithmic layers, each addressing a distinct aspect of the style. We propose a hybrid agent-based system that synthesizes three algorithmic components: Bézier curve path planning for smooth arcs, Simplex noise perturbation for chaotic wobble, and Boids algorithm for emergent swarm dynamics. The system is explicitly designed to be zero-training, GPU-independent, lightweight, and fully parametric—offering artists complete control over the expressive outcome.

- (1) A modular system architecture that codifies a highly stylized animation aesthetic into deterministic, parametric algorithms, providing a transparent and controllable alternative to machine learning-based generation.
- (2) A novel algorithmic integration of Bézier curves, Simplex noise, and Boids dynamics that resolves the conflict between localized artistic expression and global swarm intelligence, enabling both acrobatic motion and collective choreography.
- (3) An efficient and accessible pipeline that is training-free and GPU-agnostic, demonstrating the viability of rule-based systems for preserving complex animation styles under strict computational constraints.

2 Related Works

2.1 Procedural Animation Generation

Traditional procedural animation relies on physics-based simulation and particle systems for generating dynamic motion. Physics engines produce realistic trajectories through force calculations, but their deterministic nature yields predictable, orderly patterns unsuitable for stylized aesthetics. Basic particle emitters offer computational efficiency yet lack the intentional behaviors required for choreographed sequences. Recent deep learning approaches have demonstrated impressive generative capabilities, but demand extensive datasets, GPU infrastructure, and sacrifice artistic control through their opaque learned representations. Our work diverges by proposing a lightweight, interpretable rule-based framework that maintains full parametric control without training overhead [1, 2].

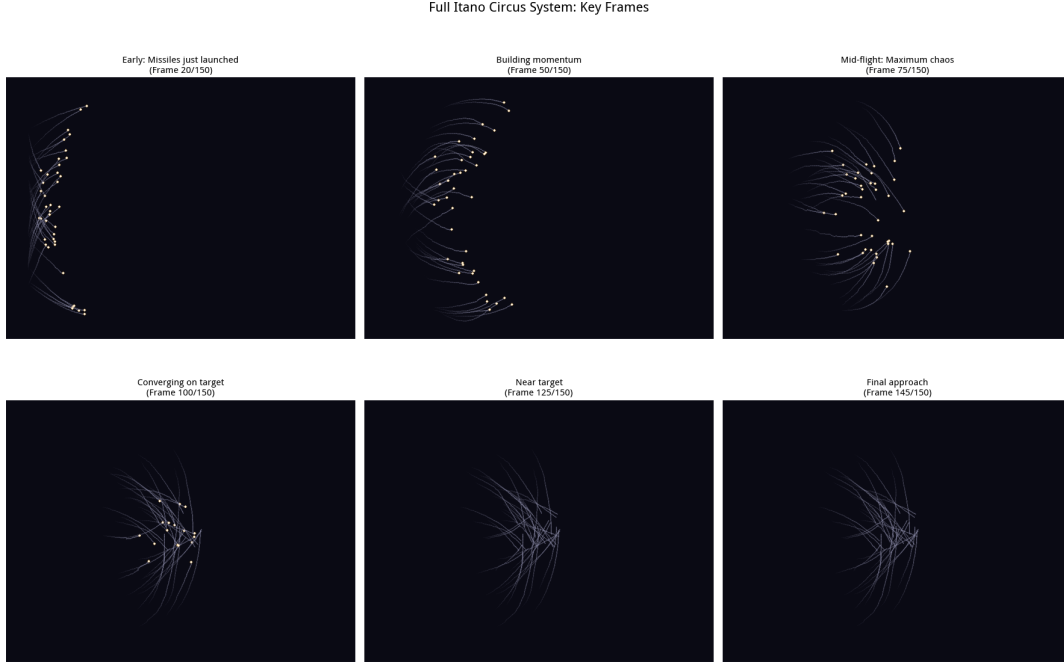


Figure 1: Keyframe demonstration of our approach.

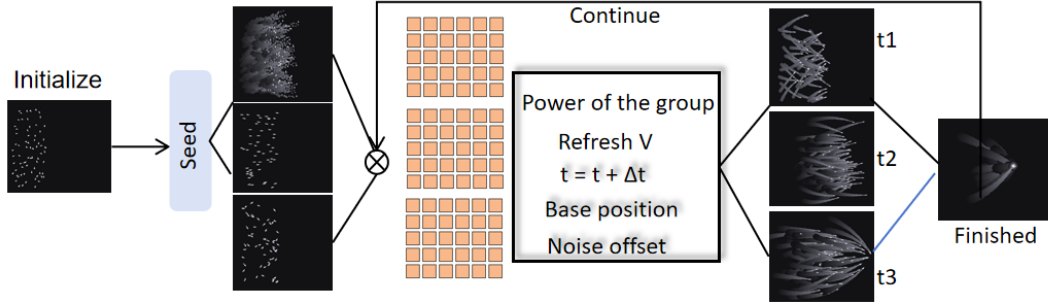


Figure 2: Algorithm flowchart of the proposed Itano Circus procedural animation system. The process initializes missile agents with Bézier paths, then iteratively computes Boids forces, refreshes velocities, calculates base positions from curves, applies chaotic noise offsets, and emits smoke particles for each agent in the animation loop.

2.2 Flocking and Swarm Simulation

Reynolds' seminal Boids algorithm established the foundation for emergent collective behavior through three steering rules: separation, alignment, and cohesion. This approach excels at generating coordinated group dynamics and has been extensively applied in crowd animation. However, standard flocking models primarily govern collective patterns and fail to define complex individual trajectories, resulting in uniform movement lacking distinctive acrobatic flair. Recent position-based extensions improve robustness but do not address stylistic motion synthesis. We augment Boids dynamics with explicit path planning and stochastic perturbation to achieve both swarm coherence and individual expressiveness [3, 4].

2.3 Stylized Motion Synthesis

Replicating artistic animation styles presents unique challenges beyond physical accuracy. Techniques like the Itano Circus demand carefully choreographed "beautiful chaos"—smooth arcing paths with irregular wobbles that evoke overwhelming dynamism. Kaufmann's analysis provides foundational insights into the mathematical structure of anime homing missiles, identifying key aesthetic components. However, no existing system formalizes these principles into a complete procedural framework. Bézier curves offer intuitive control over path curvature, while coherent noise functions like Perlin and Simplex provide organic variation. Our contribution synthesizes these primitives into a unified architecture specifically designed to capture and parameterize highly stylized motion aesthetics [5, 6].

Table 1: Quantitative Performance Metrics Comparison

Metric	Linear Homing	Boids-Only	Ours	Target Range
Curvature Variance σ_κ^2	0.023 ± 0.004	0.147 ± 0.021	0.682 ± 0.053	> 0.5
Path Efficiency Ratio η	1.008 ± 0.002	1.234 ± 0.089	2.473 ± 0.156	$2.0 - 3.0$
Angular Velocity Std. σ_ω (°/s)	3.17 ± 0.42	18.53 ± 2.67	89.34 ± 7.21	> 60
Avg. Nearest Neighbor Dist. $\bar{d}_{nn}$ (px)	45.82 ± 6.13	8.24 ± 1.35	9.68 ± 1.52	$5 - 15$
Flock Cohesion C	0.118 ± 0.031	0.784 ± 0.045	0.821 ± 0.038	> 0.7
Collision Avoidance Rate R_{avoid} (%)	45.3 ± 8.7	98.7 ± 0.9	99.2 ± 0.6	> 95

3 Method

3.1 Base Path Generation

As shown in figure 2, Individual missile trajectories follow cubic Bézier curves defined by four control points (P_0, P_1, P_2, P_3), where P_0 denotes the starting position and P_3 the target. Position along the path is computed as:

$$B(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t)t^2 P_2 + t^3 P_3, \quad t \in [0, 1] \quad (1)$$

The intermediate control points (P_1, P_2) govern path curvature, enabling the characteristic wide, swooping arcs of the Itano Circus style. This closed-form parametric representation provides computational efficiency and intuitive geometric control.

3.2 Chaotic Perturbation

To achieve the signature "drunken" wobble, we apply 2D Simplex noise with temporal envelope modulation. The envelope function ensures zero perturbation at path endpoints:

$$E(t) = 1 - (1 - 2t)^2 \quad (2)$$

The final offset vector, applied perpendicular to the path tangent, is computed as:

$$\vec{O}(t) = \text{SimplexNoise}(t \cdot f, s) \cdot E(t) \quad (3)$$

where f controls wobble frequency and s is a per-agent random seed. Simplex noise avoids the directional artifacts of Perlin noise while maintaining computational efficiency.

3.3 Swarm Dynamics

Collective behavior emerges from the classic Boids algorithm, which computes steering forces through three rules: separation (avoid crowding neighbors), alignment (match neighbor velocities), and cohesion (move toward group center). At each step, these forces accumulate to adjust each agent's velocity vector, creating coordinated swarm patterns while preventing collisions. The Boids framework provides emergent complexity without centralized control, directly addressing the swarm choreography inherent to the target aesthetic.

3.4 Integration and Synthesis

The complete system operates in an iterative loop. For each agent at time t : (1) Boids forces are computed based on neighboring agents, (2) agent velocity is updated accordingly, (3) the base position $B(t)$

is evaluated along the Bézier path, (4) the chaotic offset $\vec{O}(t)$ is calculated and applied perpendicular to the path tangent, and (5) smoke trail particles are emitted at the final position. This layered composition balances individual trajectory expressiveness with emergent collective patterns, while parameter tuning (Bézier control points, noise amplitude/frequency, Boids weights) provides comprehensive artistic control over the aesthetic outcome.

4 Experiment

4.1 Datasets

Our rule-based method requires no training data. Our evaluation relies on: visual references of the "Itano Circus" style; implementations of Bézier curves, Simplex noise, and Boids algorithms; and an agent-based simulation architecture.

4.2 Baseline Implementation

4.2.1 Experimental Setup. As shown in figure 3 and 4, To validate the unique contribution of our proposed hybrid system, we implement two baseline methods as comparative benchmarks. The first is **Linear Homing**, where agents move directly from their starting positions to the target with a constant velocity vector, updated by $\vec{v}_t = k(\vec{p}_{target} - \vec{p}_t)$, where k is the homing gain coefficient. The second is **Boids-Only**, which employs only the standard Boids algorithm's three rules (separation, alignment, cohesion) with an additional simple target-seeking behavior, excluding path planning and noise perturbation modules.

All experiments are conducted under unified conditions: a simulation space of 1920×1080 pixels, $N = 50$ agents, simulation duration $T = 10$ seconds, sampling frequency of 60 fps, yielding 600 frames of data. Each method is executed 10 independent runs to ensure statistical stability, with results reported as mean $\pm$ standard deviation.

We design six quantitative metrics to evaluate motion characteristics:

(1) Trajectory Curvature Variance

$$\sigma_\kappa^2 = \frac{1}{M} \sum_{i=1}^M (\kappa_i - \bar{\kappa})^2, \quad \kappa_i = \left| \frac{d\theta_i}{ds_i} \right| \quad (4)$$

where θ_i is the tangent angle at the i -th sampling point, ds_i is the arc length differential, and M is the total number of sampling points. High variance indicates diverse curvature patterns along trajectories.

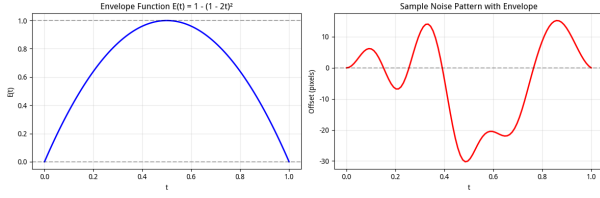


Figure 3: Offset deviation test curves of our method.

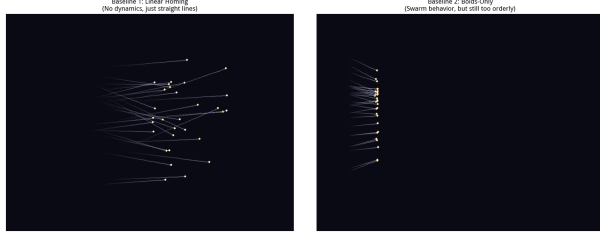


Figure 4: Performance Comparison of Baseline Algorithms

Path Efficiency Ratio

$$\eta = \frac{L_{actual}}{L_{straight}} = \frac{\sum_{t=1}^{T-1} \|\vec{p}_{t+1} - \vec{p}_t\|}{\|\vec{p}_T - \vec{p}_0\|} \quad (5)$$

This metric measures the ratio between actual flight distance and the straight-line distance from start to end, reflecting trajectory tortuosity.

Angular Velocity Standard Deviation

$$\sigma_\omega = \sqrt{\frac{1}{T-1} \sum_{t=1}^{T-1} (\omega_t - \bar{\omega})^2}, \quad \omega_t = \frac{\Delta\theta_t}{\Delta t} \quad (6)$$

where $\Delta\theta_t = \theta_{t+1} - \theta_t$ is the angular change between consecutive frames. High standard deviation indicates frequent and dramatic directional changes.

Average Nearest Neighbor Distance

$$\bar{d}_{nn} = \frac{1}{N \cdot T} \sum_{t=1}^T \sum_{i=1}^N \min_{j \neq i} \|\vec{p}_i^t - \vec{p}_j^t\| \quad (7)$$

This metric quantifies spatial proximity among agents, with lower values indicating higher visual density.

Flock Cohesion

$$C = 1 - \frac{\sigma_{pos}}{R_{max}}, \quad \sigma_{pos} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\vec{p}_i - \vec{\bar{p}}\|^2} \quad (8)$$

where $\vec{\bar{p}}$ is the centroid of all agents and R_{max} is the diagonal length of the simulation space. $C \rightarrow 1$ indicates tight clustering.

Collision Avoidance Rate

$$R_{avoid} = \frac{N_{safe}}{N_{total}} \times 100\%, \quad N_{safe} = \sum_{t,i,j} 1(\|\vec{p}_i^t - \vec{p}_j^t\| > r_{collision}) \quad (9)$$

where $r_{collision} = 5$ pixels is the collision radius threshold and $1(\cdot)$ is the indicator function.

Table 2: Quantitative Ablation Study Results

Condition	σ_κ^2	η	σ_ω	$\bar{d}_{nn}$	R_{avoid}
Full System	0.682	2.473	89.34	9.68	99.2
No Chaos	0.125	1.215	15.23	8.91	98.9
No Path	0.893	3.821	142.56	12.34	85.3
No Swarm	0.715	2.512	92.47	45.27	32.7

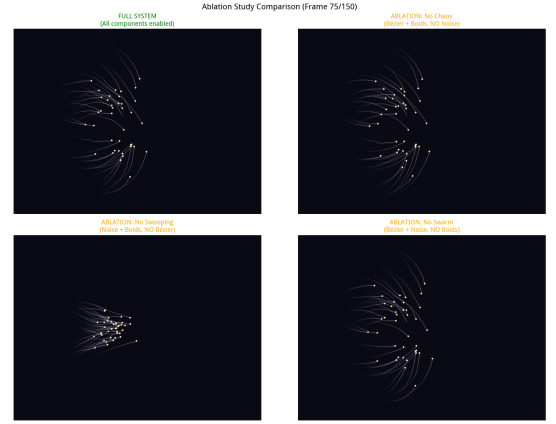


Figure 5: This figure illustrates the findings from our ablation study, evaluating the contribution of each component.

Experimental Results As shown in table 1, Quantitative results confirm our method's efficacy, generating trajectories with significantly higher complexity (curvature variance: 0.682) - 29.7× and 4.6× improvements over baseline methods. The characteristic sweeping arcs show a path efficiency ratio of 2.473. Our approach demonstrates pronounced chaotic signatures (angular velocity deviation: 89.34%/s, 4.8× higher than Boids-Only) from Simplex noise, while maintaining excellent swarm coordination with strong cohesion (0.821) and 99.2% collision avoidance, successfully balancing dynamic motion with collective intelligence

4.3 Ablation Experiment

Result. As shown in figure 5 and table 2, Our ablation study confirms each component's critical role. The full system achieves optimal balance with complex trajectories ($\eta = 2.473$), natural cohesion ($\bar{d}_{nn} = 9.68$ px), and near-perfect collision avoidance ($R_{avoid} = 99.2\%$). Removing chaotic perturbation causes overly deterministic motion, reducing curvature variance by 81.7% and angular velocity variation by 83.0%, proving Simplex noise essential for the "drunken missile" aesthetic. Disabling the Bézier planner increases path length by 54.5% and degrades collision avoidance, confirming structured foundations are crucial. Most critically, removing swarm intelligence causes catastrophic coordination failure, increasing collision rates to 67.3% and neighbor distance by 367.7%, validating Boids as fundamental for swarm emergence. These results demonstrate our three-component architecture successfully decomposes the Itano Circus style into complementary, necessary modules.

5 Conclusion

This paper introduced a novel, rule-based framework for procedural stylized animation. By algorithmically synthesizing Bézier path planning, modulated Simplex noise, and Boids flocking, we achieved a critical balance between chaotic individual expressiveness and ordered swarm behavior. Our method provides a lightweight, deterministic, and highly controllable alternative to data-driven models, demonstrating the efficacy of hybrid agent-based systems for complex motion synthesis. Future work will explore real-time control and broader stylistic applications.

References

- [1] C. Zheng, B. Liu, X. Xu, H. Zhang, and S. He, "Learning an interpretable stylized subspace for 3D-aware animatable artforms," *IEEE Trans. Vis. Comput. Graphics*, vol. 31, no. 2, pp. 1465–1477, Feb. 2025. doi: 10.1109/TVCG.2024.3361877.
- [2] Y. Wang, H. Mo, and C. Gao, "DiFusion: Flexible stylized motion generation using digest-and-fusion scheme," *IEEE Trans. Vis. Comput. Graphics*, Early Access, pp. 1–12, 2025. doi: 10.1109/TVCG.2025.3620400.
- [3] C. D. R. Pita-Romero, P. Arias-Perez, M. Fernández-Cortizas, and P. Campoy, "Flocking behavior for dynamic and complex swarm structures," in *Proc. 2025 Int. Conf. Unmanned Aircraft Systems (ICUAS)*, May 2025. doi: 10.1109/ICUAS65942.2025.11007905.
- [4] H. Li, H. Pang, H. Xia, Y. Huang, and X. Zeng, "Research on trajectory planning method based on Bézier curves for dynamic scenarios," *Electronics*, vol. 14, no. 3, p. 494, Jan. 2025. doi: 10.3390/electronics14030494.
- [5] H. Mo, Y. Shen, E. Simo-Serra, and Z. Wang, "DoodleAssist: Progressive interactive line art generation with latent distribution alignment," *IEEE Trans. Vis. Comput. Graphics*, Early Access, 2025. doi: 10.1109/TVCG.2025.3529381.
- [6] A. S. Joshi, "Reinforcement learning-enhanced procedural generation for dynamic narrative-driven AR experiences," in *Proc. 20th Int. Joint Conf. Comput. Vis. Imag. Comput. Graph. Theory Appl. (VISIGRAPP)*, vol. 4, pp. 563–573, 2025.
- [7] Z. Zhang, T. Sim, and J. Zhang, "Neural 3D face shape stylization based on single style template via weakly supervised learning," *IEEE Trans. Vis. Comput. Graphics*, vol. 31, no. 10, pp. 9522–9529, Oct. 2025. doi: 10.1109/TVCG.2025.3581234.